



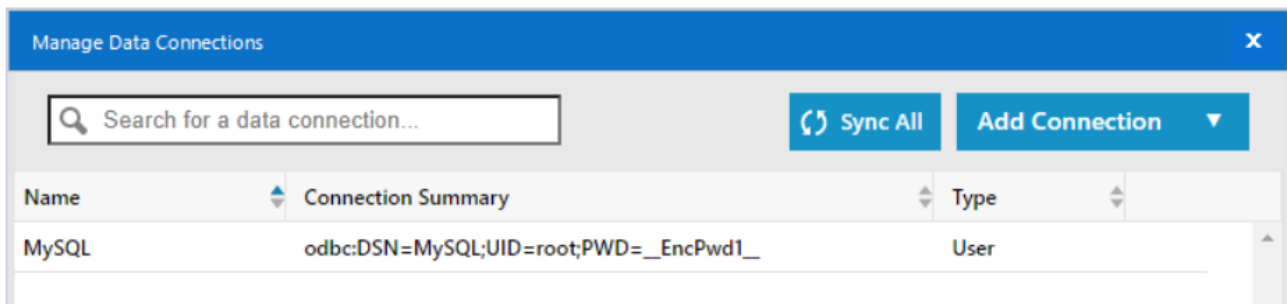
Instructions

For your group assignment, you will complete five ETL challenges. Each challenge has a series of subtasks. We have not necessarily covered the approaches needed in class, but Alteryx has great documentation to support your team.

You will be working with an Alteryx file called Medicare Claims Data (MCD), which has been provided on the group discussion page in Canvas. The file contains data from several hospitals. Each row documents a claim for a patient that was admitted to the hospital (thus patient and hospital information is repeated if they appear in more than one claim). We are particularly interested in patients that were readmitted to the hospital in 30 days or less. You can read about why here: [Hospital Readmissions Reduction Program \(HRRP\)](#).

IMPORTANT:

1. Make sure that your MySQL connection is called MYSQL



2. Do not change the names of any provided files
3. All tasks must be completed with the Alteryx tool
4. **Please do not zip your files**

Task 1: Data Conversions

1. Read **Medicare Claims Data**.
2. Label fields with the correct data type.
3. Convert all Total_Account fields to a number field.
4. Convert all dates to date data type.
5. Write out resulting data to an Alteryx file called **Task1Output**.



Task 2: Transformations and Lookups

1. Read **Task1Output**.
2. Calculate patient age (using today's date).
3. Calculate Length of Stay (LOS) for a claim.
4. Fill in any missing values for patient age (you choose what to fill in).
5. Use the ICD Diagnosis and Procedure look up spreadsheets provided to create Primary Procedure Description, Primary Diagnosis Description, Admit Diagnosis Description (use long description).
 - a. There may be some unmatched or missing codes; deal with them as you see fit.
6. Remove all unnecessary columns you may have created.
7. Write out resulting data to an Alteryx file called **Task2Output**.

Task 3: Label 30-Day Readmission Claims

1. Read **Task2Output**.
2. Create a sequence number for each individual patient claim (you will need to sort by patient ID and admit date first).
3. Create a column that indicates that the patient has been readmitted (the sequence number>1).
4. Calculate the time between readmission (hint: use the multi-row tool), and name it **ReadmitDelta**
5. Investigate the values for **ReadmitDelta**—there will be some problematic values, since this is claims data and some rows are possible accounting corrections. Find ways to remedy these data quality problems as you see fit.
6. Create a column that indicates whether the claim is a readmission for a patient that was readmitted after less than or equal to 30 days.
7. Write out resulting data to an Alteryx file called **Task3Output**.

Task 4: Create Lists of Patients, Hospitals*

1. Read **Task3Output**.
2. Patient



- a. Select attributes that are specific to a patient, and create a list where each patient is listed only once. Do not include: Admission and Discharge Date, LOS, Patient in ICU Flag, Diagnosis or Procedure Descriptions, Readmission Information.
 - b. Sort by Patient Last Name.
 - c. Create a new column that creates a bin for age (you choose the size of the bins). Make sure that you rename the bin values (for example 20-30).
 - d. Remove all unnecessary columns you may have created.
3. Create a list of unique hospitals/department combinations with all the data related to hospital (include department information); sort by hospital name.

*Note: No files will be written out when completing this task.

Task 5: Summarize Readmission Rates and R30 Rates by Hospital/Department

1. Read **Task3Output**.
2. For this task, you should have the following columns:
 - Hospital Name
 - Department Name
 - Number of Admissions (which is number of claims)
 - Sum of all Readmissions
 - Sum of R30 Readmissions
 - Readmission Rate—Sum of all Readmissions/Number of Admissions
 - R30 Rate—Sum of R30 Readmissions/Number of Admissions

Submission Instructions

- Do not rename any of the files you downloaded for this assignment.
- Put all the files in the same folders so you can use relative paths
- One team member will submit **five** workflow Alteryx files (one workflow per task).
- Please do not zip files before submitting.